Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/09/2005
PI Name:	Berg, Robert F. Ph.D.		
Project Title:	Critical Viscosity of Xenon 2 (CVX-2)		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	GOVERNMENT	**Phone:**	301-975-2466
Organization Name:	National Institute of Standards and Technology		
PI Address 1:	Process Measurements Division		
PI Address 2:	100 Bureau Drive		
PI Web Page:	http://microgravity.grc.nasa.gov/		
City:	Gaithersburg	**State:**	MD
Zip Code:	20899-8364	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	02/01/1999	**End Date:**	04/20/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Zimmerli, Greg	**Contact Phone:**	216-433-6577
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Flight Program:			
Flight Assignment:	note: extended 1 year		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Moldover, Michael (National Institute of Standards and Technology) Yao, Minwu (NASA Glenn Research Center) Zimmerli, Gregory (NASA Glenn Research Center)		
Grant/Contract No.:	C32966J		
Performance Goal No.:	1.24		
Performance Goal Text:	To Explore the Frontiers of Physics and Unlock Mysteries of the Universe Obscured by Earth’s Gravity.		

Task Description:	Base the design of CVX (Critical Viscosity of Xenon)-2 on the oscillator, sample cell, thermostat, and flight canisters used for CVX. Ground tests and the CVX flight experiment demonstrated that the performance of these components was adequate for the requirements of CVX-2. Modify the CVX electronics to allow single-frequency operation of the oscillator at amplitudes exceeding those used for CVX by as much as a factor of 30. Calculate the hydrodynamic force expected for the transverse oscillations of a cylinder of appropriate cross section. Include the effects of nonzero Reynolds number, viscoelasticity, and shear rate dependence of viscosity. Combine the results with the low-amplitude model for CVX to create a high-amplitude model for CVX-2. In microgravity, measure the response of the oscillator immersed in xenon at critical density and near the critical temperature. Use the high-amplitude model for CVX-2 to infer shear thinning by comparing the responses recorded at high oscillator amplitude with those recorded at low amplitude. The flight experiment's objective is to measure the shear thinning that is predicted to occur near the critical point of a pure fluid. Near the liquid-vapor critical point, a fluid's microscopic fluctuations become larger and slower. Their larger size causes a small increase in the viscosity. Their slower relaxation causes the viscosity to decrease if the shear rate exceeds the relaxation time. The latter effect, called shear thinning, is observed routinely in fluids with slow relaxation times such as polymer solutions. It has never been observed in a simple pure fluid such as xenon. Such a measurement cannot be done on Earth because normal gravity causes xenon's density to vary with height in the sample. The resulting inhomogeneity causes a significant decrease of the correlation length in all but an inaccessibly thin region of the sample. The range and precision of CVX-2's data will provide an accurate test of theory. The interpretation of CVX-2's data will require an accurate model of the viscometer. Creating the model is expected to lead to insights into the hydrodynamics of oscillatory flow in a previously unexplored range of Reynolds number and Keulegen-Carpenter number.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The CVX-2 experiment was a clear success. The apparatus worked well, and our analysis indicates that the data will answer the scientific question: Can a fluid as simple as xenon exhibit shear thinning? Our pleasure has been tempered by the loss of Space Shuttle Columbia and its crew, but we are grateful for the scientific opportunity given to us by STS-107. CVX-2 operated for 370 hours, probably the longest of all of Space Shuttle Columbia’s experiments. During the mission, 85 % of the data were downlinked for real time analysis. Months later, the fortunate discovery of the hard disk drive in relatively good condition increased that fraction to 99 %. The scientific analysis compares the microgravity data to preflight ground data and to the results of numerical hydrodynamic simulations and analytical calculations. Both the microgravity and ground data show that increasing the oscillator displacement increased the force/displacement ratio by as much as 20 %. Considerable effort was expended to understand this noncritical increase, which was due to the nonlinear term of the Navier-Stokes hydrodynamic equation. Dividing out the noncritical increase revealed a decrease of the force/displacement ratio that occurred only near the critical temperature (Tc) in microgravity. The relative decrease of the force/displacement ratio near Tc is small, less than 1 %. However, most of the data have a decrease that depends only on a reduced shear rate defined by the dimensionless product of the xenon’s fluctuation relaxation time and the oscillator’s maximum shear rate. This dependence qualitatively demonstrates shear-thinning near the critical point of xenon. We are working toward a quantitative understanding of the critical decrease. Efforts include numerical simulations based on the Giesekus constitutive equation and analytical calculations based on the generalized Newtonian fluid. We are working also to understand the data at higher frequencies, whose critical decrease depends on frequency as well as shear rate.
Bibliography Type:	Description: (Last Updated: 03/08/2019)
Articles in Peer-reviewed Journals	Berg RF. "Fluids near a critical point obey a generalized Cox–Merz rule." Journal of Rheology 48, 1365-73. https://ws680.nist.gov/publication/get_pdf.cfm?pub_id=903224 , Nov-2004